

Reference Guide — Dashboards 1–3 (Customer Support, Retail Sales, Finance Operations)

This document is a complete A–Z reference of the three Excel dashboards we built together: Dashboard 1 (Customer Support Operations), Dashboard 2 (Retail Sales & Business Performance), and Dashboard 3 (Finance Operations). Each section includes dataset notes, full cleaning steps, helper columns and formulas, pivot and visual construction, slicer configuration, troubleshooting notes, and final layout/polish instructions. For each formula or step the exact cell/table references and rationale are included so you can reproduce, audit, or hand off the workbook.

How to use this guide

Structure: Each dashboard section is organized as **Dataset & original schema**, **Data audit & cleaning**, **Helper columns & formulas**, **Pivot tables and charts**, and **Dashboard assembly & polish**. Use the table of contents to jump to a specific area or reproduce the workbook step-by-step.

Dashboard 1 — Customer Support Operations

Dataset & initial inspection

Source: Kaggle-style Customer Support Ticket dataset. Typical columns observed: Ticket ID; Date Created / Date Resolved; Agent Name; Team / Shift; Channel; Ticket Type; Priority; Status; Resolution Time (Hrs); CSAT Score; SLA Breached; First Contact Resolution. Row count used in practice: 8,469.

First-step audit checklist (what to run/look for manually): row/column counts, date range, unique values in Status and Channel, visible blanks in CSAT or Resolution Time, inconsistent casing/spaces in text columns, duplicate Ticket ID check.

Governance: sheet layout and conventions

- Raw_Data: untouched original CSV copy
- Working_Data: active table for cleaning and helper columns
- Audit_Check: small calculations (COUNTBLANK, COUNTA) and diagnostics
- KPI_Summary: all KPI formulas (single source of truth for cards)
- Dashboard: final polished layout with KPI cards, charts, and slicers

Cleaning steps (exact actions and formulas)

1. Convert raw data to an Excel Table (select anywhere in data → Ctrl+T). Rename table to CustomerSupport for stable structured references.
2. Add Row_Check column to working table to give each record a stable row number: header Row_Check, formula in row 2: =ROW()-1. This preserves traceability when sorting.
3. Check duplicates on Ticket ID: select Ticket ID column → Conditional Formatting → Highlight Cell Rules → Duplicate Values. Note duplicates found: None in practice. If duplicates present, inspect and decide deduplicate or merge.
4. Fix Resolution casing inconsistency: add resolution clean column header Resolution_Clean; formula: =PROPER([@Resolution]) — this standardizes SOLVED/solved/Solved into Solved.

- Trim name fields to remove invisible spaces: add Customer_Name_Clean, formula =TRIM([@[Customer Name]]).
- Count blanks in CSAT using Audit_Check sheet (use Table references to avoid entire-column counts). Example: =COUNTBLANK(CustomerSupport[Customer Satisfaction Rating]) and =COUNTA(CustomerSupport[Customer Satisfaction Rating]). In practice: 2,769 filled and 5,700 blanks; blank % = 67.3% — flag this for any CSAT averages.
- Add month/year helper columns to support time grouping: Month_Number =MONTH([@[Date of Purchase]]); Month_Name =TEXT([@[Date of Purchase]],"mmm"); Year =YEAR([@[Date of Purchase]]). If TEXT formula errors, check date cell types; use direct cell reference as fallback.

KPI summary (formulas used)

KPI	Formula (example)
Total Tickets	=COUNTA(CustomerSupport[Ticket ID])
Open Tickets	=COUNTIF(CustomerSupport[Status],"Open")
Closed Tickets	=COUNTIF(CustomerSupport[Status],"Closed")
Pending Tickets	=COUNTIF(CustomerSupport[Status],"Pending")
Average CSAT (note: only filled rows)	=AVERAGE(CustomerSupport[Customer Satisfaction Rating])

Pivot tables and charts

Pivot 1: Tickets by Status — Rows: Status; Values: Count of Ticket ID. Pivot 2: Tickets by Category/Type — Rows: Ticket Type; Values: Count of Ticket ID. Charts: pie/doughnut for status distribution; bar chart for category volumes. Top 3 categories observed: Refund Request, Technical Issue, Cancellation Request. Highest-status count observed: Pending (backlog risk).

Trend / performance chart attempt

We explored average resolution time trend using First Response Date and Time to Resolution. Helper column 'Resolution Time (Hours)' used: =IF(OR([@[First Response Date]]="",[@[Time to Resolution]]=""),"",[@[Time to Resolution]]-[@[First Response Date]]*24). Built pivot (First Response Date grouped by Month → Average of Resolution Time) and line chart. Decision: omitted from final dashboard because the visual did not meaningfully communicate performance for this dataset state.

Dashboard assembly & interactivity

- Create Dashboard sheet. Add KPI cards at top: merge small cell blocks for each card, link to KPI_Summary values (e.g., =KPI_Summary!B2). Style: large bold number, label above, light fill.
- Paste charts onto Dashboard sheet and position under KPI row.
- Insert slicers (PivotTable Analyze → Insert Slicer) for Status and Ticket Type. Connect slicers to all pivot tables via Slicer → Report Connections so selecting a slicer value filters every visual at once.
- Final polish: remove gridlines (View → Gridlines), set consistent color palette across charts, hide backend sheets (right-click tab → Hide), protect sheet if required.

Troubleshooting & lessons learned

Common pitfalls encountered and solutions:• Pivot caches: If two pivots were built from different ranges or created separately, a slicer may only connect to the pivot it was created from. Fix: rebuild the second pivot from the same named Table or copy the pivot as a pivot object ensuring it uses the same cache.• COUNTBLANK/COUNTA must reference Table ranges to avoid counting entire Excel column blanks

(millions of rows). Use structured references: CustomerSupport[Customer Satisfaction Rating].• Text normalisation matters for grouping: use PROPER, UPPER, TRIM to standardise before pivoting.

Dashboard 2 — Retail Sales & Business Performance

Dataset & initial checks

Source: Superstore-like dataset (Order ID, Order Date, Ship Date, Customer, Region, Product, Sales, Quantity, Discount, Profit). The dataset used had mixed date formats and required careful date cleaning before building time-based pivots.

Date cleaning (mixed formats)

Symptoms: some cells parsed as dates (ISNUMBER returns TRUE), others remained text (ISNUMBER returns FALSE). Example formats detected: 12-05-2016 (interpreted as DD-MM-YYYY depending on system) and 11/22/2015 (MM/DD/YYYY).

Fix approach: use a helper column and a DATE/DATEVALUE reconstruction. Example safe helper formula for a cell A2:

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=IF ( ISNUMBER ( A2 ) , A2 , IF ( ISNUMBER ( SEARCH ( " - " , A2 ) ) , DATE ( VALUE ( RIGHT ( A2 , 4 ) ) , VALUE ( LEFT ( A2 , 2 ) ) , VALUE ( LEFT ( A2 , 2 ) ) ) , DATE ( VALUE ( LEFT ( A2 , 2 ) ) , VALUE ( RIGHT ( A2 , 2 ) ) , VALUE ( RIGHT ( A2 , 2 ) ) ) ) ) )
```

Validate results using ISNUMBER on the helper column and then Paste → Values over the original date column. Repeat for Ship Date if needed.

Lookup techniques — VLOOKUP, XLOOKUP, INDEX+MATCH

Task: enrich sales table with Regional Manager based on Region. A small RegionManager lookup table was created (Region → Manager). Three lookup techniques were demonstrated to compare pros/cons.

Technique	Example formula	Notes
VLOOKUP	=VLOOKUP([@Region],RegionManager,2,FALSE)	Classic. Works left-to-right only; sensitive to extra spaces or spelling mismatches; wrap lookup value with TRIM to avoid whitespace issues.
XLOOKUP	=XLOOKUP([@Region],RegionManager[Region],RegionManager[Regional Manager],"Not Found")	Modern and preferred; supports any direction, explicit if_not_found argument; robust and easier to read.
INDEX+MATCH	=INDEX(RegionManager[Regional Manager],MATCH([@Region],RegionManager[Region],0))	Flexible and still useful for legacy compatibility; MATCH returns position that INDEX uses; must lock lookup range with \$ for fill-down.

KPI cards and pivot visual building

1. Create Dashboard sheet and layout 4 KPI cards in a row: Total Sales, Total Profit, Total Orders, Average Order Value. Example formulas: =SUM(SalesRange), =SUM(ProfitRange), =COUNTA(OrderIDRange), AvgOrderValue = TotalSales / TotalOrders.
2. Build Pivot: Sales by Region — Region to Rows; Sales to Values (Sum). Create PivotChart → Clustered Column, move to Dashboard.
3. Build Pivot: Sales Trend — Order Date to Rows; Group by Months (right-click → Group); Sales to Values (Sum); PivotChart → Line → move to Dashboard.

4. Build Pivot: Top Products — Product to Rows; Sales to Values; Value Filters → Top 10 by Sum of Sales; PivotChart → Bar.
5. Build Pivot: Profit Margin — add helper column Profit Margin = Profit / Sales (if desired), or use Pivot calculated field: ProfitMargin% = Sum(Profit)/Sum(Sales) (via calculated field or using two pivot values and a chart).

Slicers, cache & connections

Slicers were added for Region, Manager, and Month. Important technical note: when multiple pivot tables are used on the same sheet, they must share the same pivot cache to be connectable by the same slicer. If a pivot was copied incorrectly and built on a distinct cache, the slicer won't list it in Report Connections. Fix: recreate the second pivot by creating it from the same named Table or copy the pivot object using PivotTable Analyze → Select → Entire PivotTable, then paste; ensure both pivots reference the same Table name.

Dashboard polish

- Title banner + subtitle.
- Row with slicers above KPI cards.
- KPI cards styled consistently: bold numeric, muted label, light fill.
- 2×2 chart grid aligned and sized identically (use Format → Size to set exact dimensions).
- Hide backend Pivot sheets and raw data sheets to deliver a clean presentation.

Dashboard 3 — Finance Operations (AP + AR combined)

Project goal and datasets

Goal: build a single Finance Operations dashboard combining Accounts Payable (AP) and Accounts Receivable (AR) so the organization can see outstanding balances, aging, disputed amounts, and late-payment trends. Sources used: an AP dataset (APID, Vendor, InvoiceDate, DueDate, Amount, Currency, Status, PaidDate, Terms) and an IBM AR dataset (countryCode, customerID, PaperlessDate, invoiceNumber, InvoiceDate, DueDate, InvoiceAmount, Disputed, SettledDate, PaperlessBill, DaysToSettle, DaysLate).

Unified schema and reasoning

To stack AP and AR into one table we defined a neutral schema: TransactionID, TransactionType (Payable/Receivable), PartyName (Vendor/Customer), InvoiceDate, DueDate, Amount, Currency, PaymentStatus, PaidDate, DaysLate, DisputedFlag, Country, PaperlessDate, PaperlessBill, Terms, DatasetSource. This preserves traceability and allows combined pivots and cross-type KPIs.

Step-by-step transformation (AP then AR)

1. AP_Raw: clean date columns (InvoiceDate, DueDate, PaidDate) using the date helper formula approach and convert to real Dates. Add DaysLate_Calc = IF(PaidDate<>,"MAX(PaidDate-DueDate,0)"). Format DaysLate as Number, set 0 for on-time/early. Convert formulas to Values.
2. AP_Transformed: map AP_Raw columns to the 16-column unified schema using simple cell references/formulas. Example: TransactionID = AP_Raw!A2; TransactionType = "Payable"; PartyName = AP_Raw!B2; PaidDate = AP_Raw!H2; DaysLate = AP_Raw!J2; DatasetSource = "AP". Fill down and Copy → Paste as Values.
3. AR_Raw: fix PaperlessDate, InvoiceDate, DueDate, SettledDate date formats with helper columns, convert to values. Derive PaymentStatus:

=IF(SettledDate<>"",Paid,IF(TODAY()>DueDate,Overdue,'Pending')). Ensure DaysLate column from AR was preserved and correct.

4. AR_Transformed: map AR_Raw to unified schema. Example: TransactionID = AR_Raw!invoiceNumber; TransactionType = "Receivable"; PartyName = AR_Raw!customerID; Amount = AR_Raw!InvoiceAmount; DisputedFlag = AR_Raw!Disputed; DatasetSource = "IBM_AR". Paste values to lock.
5. Raw_Data: Create final unified sheet with headers for the 16 schema columns, then copy AP_Transformed rows into Raw_Data then AR_Transformed rows immediately below. Validate counts and a few boundary rows.

Helper columns and analytic flags

Add the following columns (in Raw_Data) to make pivoting simple. Use these exact formulas in row 2 and fill down, then Paste as Values if you want to freeze them:

Column	Formula (example)	Purpose
AgingBucket	=IF(J2="",N/A,IF(J2<=0,"On Time",IF(J2<=30,"1-30 Days",IF(J2<=60,"31-60 Days",IF(J2<=90,"61-90 Days","90+ Days")))))	Categorise overdue amounts for aging charts
IsLate	=IF(J2>0,1,0)	Binary flag for late payments (1=late,0=on time)
IsDisputed	=IF(OR(K2=TRUE,K2="Yes",K2=1),1,0)	Binary disputed flag normalized across possible source values
IsOpen	=IF(H2="Open",1,0)	Open/unpaid flag based on PaymentStatus
Year	=IF(D2<>"",YEAR(D2),"")	Transaction year for trend pivots

Pivot tables, KPIs, and charts

- Pivot: Aging Summary — Rows: AgingBucket; Values: Sum of Amount, Count of TransactionID. Chart type: Column with optional line for counts.
- Pivot: Late Payment % by Year & Type — Rows: Year; Columns: TransactionType; Values: Average of IsLate (format as %). Chart: Line with markers.
- Pivot: Disputed Amount — Rows: IsDisputed; Values: Sum of Amount, Count of TransactionID. Chart: Donut/Pie showing disputed vs not disputed.
- Pivot: Open Outstanding by Type — Rows: TransactionType; Values: Sum of Amount with a Report filter or slicer set to IsOpen=1. Chart: Horizontal Bar.

KPI cards (formulas and rationale)

KPI	Formula (example)	Why it matters
Total Outstanding	=SUMIFS(Raw_Data!Amount, Raw_Data!IsOpen, 1)	Total unpaid amount across AP & AR — cash at risk / working capital exposure
Total Disputed Amount	=SUMIFS(Raw_Data!Amount, Raw_Data!IsDisputed, 1)	Value currently under dispute — requires resolution actions
% Invoices Late	=AVERAGE(Raw_Data!IsLate)	Average of 1/0 flag gives percent late — track process health

KPI	Formula (example)	Why it matters
Total Payable Balance	=SUMIFS(Raw_Data!Amount, Raw_Data!TransactionType, "Payable", Raw_Data!IsOpen, 1)	How much the company owes currently (outstanding AP)
Total Receivable Balance	=SUMIFS(Raw_Data!Amount, Raw_Data!TransactionType, "Receivable", Raw_Data!IsOpen, 1)	How much others owe the company (outstanding AR)

Dashboard layout and finalization

1. Create Dashboard sheet. Add slicers (top-left) for TransactionType, Year, AgingBucket.
2. Place KPI cards beneath slicers, then arrange 4 charts in a 2×2 grid: Aging Summary, Late Payment Trend, Disputed Amount, Open Balance comparison.
3. Connect each slicer to all pivot tables (right-click slicer → Report Connections → check all pivot tables) so filtering updates all visuals concurrently.
4. Add a small footer with: Data as of [date] | Source: AP_Raw, AR_Raw | Prepared by: [Your name]. Hide backend sheets. Turn off gridlines and freeze the title row if desired.

Practical notes, data caveats & audit signals

Key issues we explicitly documented and why: CSAT missing % in Dashboard 1 must be disclosed whenever you publish an average; in Dashboard 3 we intentionally noted any artificial edits (manually added PaidDates or country codes) as practice-only changes. In deliverable files for real clients never fabricate values; instead flag missing fields for source-data correction. Always maintain Raw_Data as an untouched copy to enable audit trails.

Appendix: Common formulas and quick references

Purpose	Formula (example)	Notes
Count blank within table	=COUNTBLANK(Table[ColumnName])	Use structured references to avoid counting entire column beyond your data.
Count non-blank within table	=COUNTA(Table[ColumnName])	Counts non-empty cells; use with the same TableName.
Days Late (AP)	=IF(PaidDate<>"",MAX(PaidDate-DueDate,0),"")	Result is days; format as Number. Blank for unpaid rows.
Aging bucket	=IF(DaysLate="",N/A,IF(DaysLate<=0,"On Time",IF(DaysLate<=30,"1-30 Days",IF(DaysLate<=60,"31-60 Days",IF(DaysLate<=90,"61-90 Days",IF(DaysLate<=90,"90+ Days")))))	Nested IF converts numeric into category.
SUMIFS multi-criteria	=SUMIFS(SumRange,CriteriaRange1,Criteria1,CriteriaRange2,Criteria2)	Each additional pair is an AND filter.
XLOOKUP example	=XLOOKUP(LookupValue, LookupArray, ReturnArray, "Not Found")	Preferred for new Excel versions; replaces VLOOKUP & INDEX+MATCH in most cases.

Purpose	Formula (example)	Notes
INDEX+MATCH	=INDEX(ReturnRange, MATCH(LookupValue, LookupRange, 0))	Useful for legacy files or special two-way lookups.
Sentence case in Power Query (M)	Text.Upper(Text.Start([title],1)) & Text.Lower(Text.End([title], Text.Length([title]) - 1))	Used in Power Query Custom Column to capitalise first letter only.

Final confirmation & next steps

This document contains the full, explicit instructions, formulas, pivot and chart definitions, and operational notes for Dashboards 1–3 as requested. It excludes Dashboard 4 / Phase 6 content per your instruction. If you would like, I can now:

- Export this content into a formatted PDF (I can generate the downloadable PDF for you if your chat environment supports the 'Generate Document' feature).
- Produce a shorter executive one-page PDF per dashboard suitable for client delivery.
- Produce a 'clean' workbook checklist that lists every cell and sheet to be present in the deliverable.

Please confirm you want me to produce the PDF now. If your interface does not display the 'Generate Document' option, tell me your preferred export method (I will provide the content in a single copyable block you can paste into Word and export as PDF).